FTEC5660 Reproducibility Work

Working Memory Capacity of ChatGPT: An Empirical Study

1155245678 XuRuoxuan

1 Project Summary + What I Tried to Reproduce

1.1 Project Summary

Background

Working memory is a critical component of both human and artificial intelligence, defined as the ability to temporarily store and manipulate information. In cognitive science, the n-back task is considered the gold standard for measuring working memory capacity. Participants must continuously monitor a stream of stimuli and respond when the current stimulus matches the one presented n steps earlier, requiring both storage and executive attention processes.

The paper "Working Memory Capacity of ChatGPT: An Empirical Study" by Gong et al. (2023) systematically evaluates ChatGPT's working memory capacity using verbal and spatial n-back tasks. The authors find that ChatGPT exhibits a human-like capacity limit, with performance declining significantly as n increases, particularly dropping when n reaches 3. They further investigate how different instruction strategies, such as adding noise, providing feedback, and chain-of-thought prompting, affect the model's performance.

This Study

This project aims to reproduce a subset of the original study's findings and extend them with a novel modification. Specifically, I focus on the verbal n-back baseline condition (no noise, no feedback, no chain-of-thought) for n = 1, 2, 3, using the same experimental parameters: 50 blocks per n, 24 trials per block (8 match, 16 nonmatch), and the same letter sequences.

After establishing the baseline, I introduce a modification by varying the model's temperature parameter—a core parameter controlling output randomness. I test three temperature settings: 0.2 (more deterministic), 1.0 (baseline, as used in the original study), and 1.5 (more random). This modification addresses a question not explored in the original paper: How does the model's internal "decision strategy" (deterministic vs. random) affect its working memory performance?

All four standard metrics—hit rate, false alarm rate, accuracy, and d'—are computed for each condition, with response time also recorded as a supplementary measure.

1.2 What I Tried to Reproduce

In this project, I focused on reproducing the baseline verbal n-back results

reported in Figure 5 of the original paper by Gong et al. This figure presents ChatGPT's performance across three levels of task difficulty (n = 1, 2, 3) under the standard condition—no noise, no feedback, and no chain-of-thought prompting. This baseline condition is fundamental to the paper's core finding that ChatGPT exhibits a human-like working memory capacity limit.

For each level of n, I used the same experimental parameters as the original study: 50 blocks per n, with each block containing 24 trials—8 match trials and 16 nonmatch trials. The letter sequences were generated from the alphabet "bcdfghjklnpqrstvwxyz", excluding vowels to minimize verbal rehearsal strategies. Each letter was presented one at a time through the OpenAI API, and the model was instructed to respond with "m" on match trials and "-" on nonmatch trials.

The model used was gpt-3.5-turbo with temperature set to 1.0, matching the original study's configuration. I recorded both the model's responses and the response time for each trial.

2 Setup Notes

2.1 Environment

OS: Windows 11
Python Version: 3.9.18 (Anaconda environment)
Key Libraries: openai, numpy, pandas, scipy, matplotlib, seaborn, json, time, random
IDE: Jupyter Notebook
Hardware: Local machine, 32GB RAM

2.2 Data

Stimuli Generation: Letter sequences generated from alphabet (21 consonants, excluding vowels to minimize verbal rehearsal)
Experimental Parameters:
Blocks per n: 50
Trials per block: 24 (8 match, 16 nonmatch)
Total trials per condition: 1,200
Data Storage: Raw responses saved in JSON format (all_trials_verbal.json); processed metrics exported to CSV for analysis

2.3 API Keys & Access

Provider: OpenAI
Model: gpt-3.5-turbo
Authentication: API key stored as environment variable OPENAI_API_KEY

3 Reproduction Target(s) + Metric Definition

3.1 Reproduction Target(s)

The primary target of my reproduction is the baseline verbal n-back performance across three levels of task difficulty: n = 1, 2, and 3 reported in Figure 5 of the original paper by Gong et al. More important than matching exact numbers is reproducing the monotonic declining trend—the finding that ChatGPT's performance systematically decreases as working memory load increases, with a particularly notable drop when n reaches 3. This pattern constitutes the core evidence for the paper's claim that ChatGPT exhibits a human-like working memory capacity limit.

3.2 Metric Definitions

To comprehensively evaluate model performance, I follow the original study in using four standard metrics from signal detection theory and behavioral research. Each metric captures a different aspect of working memory performance:

1. **Hit Rate (HR)**: Measures the model's ability to correctly identify match trials—instances where the current letter is the same as the letter presented n steps earlier.

Formula: HR = Hits / (Hits + Misses)

Interpretation: Higher values indicate better target detection.

2. **False Alarm Rate (FAR)**: Measures the model's tendency to incorrectly respond "m" on nonmatch trials—instances where no match actually exists.

Formula: FAR = False Alarms / (False Alarms + Correct Rejections)

Interpretation: Lower values indicate better inhibition of inappropriate responses.

3. **Accuracy (ACC)**: Provides an overall measure of performance across all trial types.

Formula: ACC = (Hits + Correct Rejections) / Total Trials

Interpretation: While intuitive, accuracy can be biased by imbalanced trial design (each block has 8 match vs. 16 nonmatch trials). A model that always responds "-" would achieve 67% accuracy without any working memory.

4. **d' (Detection Sensitivity)**: Combines hit rate and false alarm rate into a single bias-free measure of discriminability, derived from signal detection theory.

Formula: d' = z(HR) - z(FAR), where z is the inverse of the normal distribution function

Interpretation: Higher values indicate better ability to distinguish between targets and non-targets, independent of response bias. Values above 1.0 are generally considered above-chance performance.

4 Results: My Numbers vs Reported Numbers

I successfully collected data from 50 blocks per n-back level (1, 2, and 3) under the baseline condition (temperature = 1.0, no noise, no feedback, no chain-of-thought). The results are summarized in Table 1, alongside the original paper's reported values for comparison. All values are presented as mean ± standard error of the mean (SEM), following the original study's reporting format.

N-back	Hit Rate (%)	False Alarm Rate (%)	Accuracy (%)	D Prime
1-back				
My Results	93.00 ± 1.42	1.12 ± 0.38	96.92 ± 0.51	3.99 ± 0.10
Paper Results	87.00 ± 1.62	1.50 ± 0.42	94.67 ± 0.58	3.53 ± 0.11
2-back				
My Results	21.00 ± 2.22	7.12 ± 0.79	68.92 ± 0.84	0.83 ± 0.08
Paper Results	32.00 ± 2.13	4.12 ± 0.76	74.58 ± 1.00	1.43 ± 0.10
3-back				
My Results	14.75 ± 1.65	5.00 ± 0.73	68.25 ± 0.74	0.82 ± 0.08
Paper Results	23.25 ± 1.62	4.50 ± 0.66	71.42 ± 0.82	1.11 ± 0.09

Table 1: Comparison of Experimental Results

4.1 Comparison of Results

For 1-back, my results show higher performance across all metrics compared to the original paper. The hit rate (93.00% vs. 87.00%), accuracy (96.92% vs. 94.67%), and d' (3.99 vs. 3.53) are all notably higher, while the false alarm rate is slightly lower (1.12% vs. 1.50%). This indicates that on the easiest task, my replication achieved superior performance to what was originally reported.

For 2-back, a striking reversal occurs. My hit rate (21.00%) is substantially lower than the paper's (32.00%), while my false alarm rate (7.12%) is nearly double the paper's (4.12%). Consequently, my d' drops to 0.83—well below the paper's 1.43 and falling under the d' = 1 threshold that the paper uses to define working memory capacity. Accuracy also declines to 68.92%, barely above the trivial baseline of always responding "-" (66.7%).

For 3-back, the pattern continues with my hit rate (14.75%) below the paper's (23.25%), false alarm rate (5.00%) slightly higher than the paper's (4.50%), and d' (0.82) below the paper's (1.11). My accuracy (68.25%) remains near the trivial baseline.

The most notable finding is that in my replication, ChatGPT's working memory capacity limit appears at n = 2 rather than n = 3 as reported in the original study. The d' values at both n=2 and n=3 are below 1.0, indicating near-chance performance, whereas the original study maintained d' above 1.0 even at n=3.

4.2 Metric-by-Metric Analysis

d' (Detection Sensitivity)

My results show a clear capacity limit at n=2, with d' dropping from 3.99 (excellent) to 0.83 (near-chance). The original paper shows a more gradual decline (3.53 → 1.43 → 1.11), maintaining discriminability above chance at n=3. Since stimuli and code are identical, this difference suggests the model version I accessed may have different working memory characteristics than the one used in the original study.

Hit Rate

The hit rate reveals a steep performance cliff between n=1 and n=2 in my results (93.0% → 21.0%), compared to a more moderate decline in the paper (87.0% → 32.0%). The high standard error at n=2 (2.22) indicates considerable block-to-block variability—the model struggles inconsistently at this difficulty level.

False Alarm Rate

My false alarm rate increases dramatically at n=2 (1.12% → 7.12%), nearly double the paper's increase. This suggests that under uncertainty, my model adopts a more liberal response strategy—guessing "m" more frequently. The slight decrease at n=3 (to 5.00%) is non-monotonic and may reflect a shift back to a more conservative strategy after poor performance.

Accuracy

Accuracy confirms the pattern: at n=2 and n=3, my results (68.92%, 68.25%) are barely above the 66.7% trivial baseline of always saying "-", while the paper's accuracy remains above 70% even at n=3.

4.3 Overall Pattern and Possible Explanations

What Reproduces Well: The qualitative pattern of declining performance with increasing n is clearly reproduced, confirming the paper's core claim that ChatGPT has limited working memory capacity.

What Differs: The quantitative capacity limit appears shifted—my results show collapse at n=2, while the original paper shows measurable performance still at n=3.

Possible Explanations (since stimuli and code are identical):

Model Version Changes: OpenAI continuously updates gpt-3.5-turbo. The 2024 version I accessed may differ from the 2023 version used in the original study, potentially affecting working memory performance at higher cognitive loads.

Stochastic Variation: LLM outputs have inherent randomness even at temperature=1.0. My results may represent one end of a performance distribution.

API Infrastructure Changes: Even with identical code, underlying API routing, load balancing, or inference optimizations may have changed.

Conclusion: Despite quantitative differences, the core finding is robustly reproduced. My results actually show an even stricter capacity limit than the original paper, reinforcing that LLMs, like humans, cannot maintain unlimited information in working memory.

5 My Modification + Results After Modification

My modification to the original experimental design was a parameter change: I varied the temperature parameter of the model while keeping all other aspects identical to the baseline condition. Temperature controls the randomness of the model's output—lower values make the model more deterministic and conservative, while higher values increase randomness and exploration.

Specifically, I ran the same verbal n-back task (n = 1, 2, 3, 50 blocks each) under two additional temperature settings:

Low Temperature (0.2): More deterministic outputs, higher probability of selecting the most likely token

High Temperature (1.5): More random outputs, flatter probability distribution allowing less likely tokens to be selected

5.1 Results: Low Temperature (0.2)

N-back	Hit Rate (%)	False Alarm Rate (%)	Accuracy (%)	D Prime
1-back	92.25 ± 1.62	1.00 ± 0.37	96.75 ± 0.56	3.98 ± 0.11
2-back	19.25 ± 2.40	5.38 ± 0.75	69.50 ± 0.92	0.88 ± 0.09
3-back	9.00 ± 1.37	3.62 ± 0.59	67.25 ± 0.61	0.74 ± 0.07

Table 2: Performance at Temperature = 0.2

Metric	1-back	2-back	3-back
Hit Rate	-0.75%	-1.75%	-5.75%
False Alarm Rate	-0.12%	-1.74%	-1.38%
Accuracy	-0.17%	+0.58%	-1.00%
D Prime	-0.01	+0.05	-0.08

Table 3: Comparison with Baseline (Temperature = 1.0)

At low temperature (0.2), the model shows slightly lower hit rates across all n levels, particularly at n=3 where hit rate drops by 5.75 percentage points. However, it also shows lower false alarm rates at n=2 and n=3 (reductions of 1.74 and 1.38 percentage points respectively). This suggests a trade-off: the model becomes more conservative—it says "m" less often overall, which reduces both correct detections (hits) and incorrect alarms (false alarms).

The d' values remain very close to baseline at n=1 and n=2, but show a slight decrease at n=3 (from 0.82 to 0.74). This indicates that increased determinism does not improve the model's ability to discriminate targets from non-targets; in fact, at the highest difficulty level, it slightly impairs performance.

Interestingly, the standard errors at low temperature are generally similar to baseline, suggesting that reducing randomness does not make the model's performance more consistent across blocks.

5.2 Results: High Temperature (1.5)

N-back	Hit Rate (%)	False Alarm Rate (%)	Accuracy (%)	D Prime
1-back	91.00 ± 1.62	2.50 ± 0.59	95.33 ± 0.64	3.71 ± 0.12
2-back	27.75 ± 2.27	7.25 ± 0.94	71.08 ± 1.00	1.04 ± 0.09
3-back	21.50 ± 1.62	7.38 ± 0.86	68.92 ± 0.78	0.89 ± 0.07

Table 4: Performance at Temperature = 1.5

Metric	1-back	2-back	3-back
Hit Rate	-2.00%	+6.75%	+6.75%
False Alarm Rate	+1.38%	+0.13%	+2.38%
Accuracy	-1.59%	+2.16%	+0.67%
D Prime	-0.28	+0.21	+0.07

Table 5: Comparison with Baseline (Temperature = 1.0)

High temperature (1.5) produces the most interesting results. At n=1, performance declines across all metrics—hit rate drops, false alarm rate increases, and d' decreases by 0.28. This suggests that adding randomness to an easy task only harms performance.

However, at n=2 and n=3, a surprising pattern emerges: hit rates increase substantially (by 6.75 percentage points at both levels), and d' increases at n=2 (from 0.83 to 1.04, crossing the d'=1 threshold). False alarm rates also increase, but the net effect on d' is positive at n=2.

This indicates that at higher difficulty levels, increased randomness can actually improve performance. One possible explanation is that when the model is uncertain (as it often is at n=2 and n=3), a more random output distribution allows it to occasionally "guess" correctly when it otherwise might default to the wrong response. In other words, randomness helps overcome a systematic bias toward saying "-" at higher n levels.

The standard errors at high temperature are generally larger than baseline, particularly for false alarm rate at n=2 (0.94 vs. 0.79) and n=3 (0.86 vs. 0.73), confirming that increased randomness leads to less consistent performance across blocks.

5.3 Temperature Effects Summary

N-back	Temp=0.2	Temp=1.0 (Baseline)	Temp=1.5
1-back	3.98	3.99	3.71
2-back	0.88	0.83	1.04
3-back	0.74	0.82	0.89

Table 6: D Prime Across Temperature Conditions

The results reveal a striking pattern: **temperature affects performance differently depending on task difficulty**.

At low temperature (0.2): The model becomes more conservative, reducing both hits and false alarms. This slightly impairs performance at the highest load (n=3), suggesting that excessive determinism does not help working memory when the task is already difficult.

At high temperature (1.5): A U-shaped effect emerges—performance declines on the easy task (n=1) but improves on harder tasks (n=2 and n=3). Most notably, at n=2, d' increases from 0.83 to 1.04, crossing the d'=1 threshold. This means the model achieves above-chance performance where baseline was near-chance.

This pattern suggests that when the model is uncertain (as at higher n levels), a more random output distribution allows it to occasionally "guess" correctly, overcoming a systematic bias toward saying "-". In contrast, when the task is easy and the model is confident, adding randomness only introduces errors.

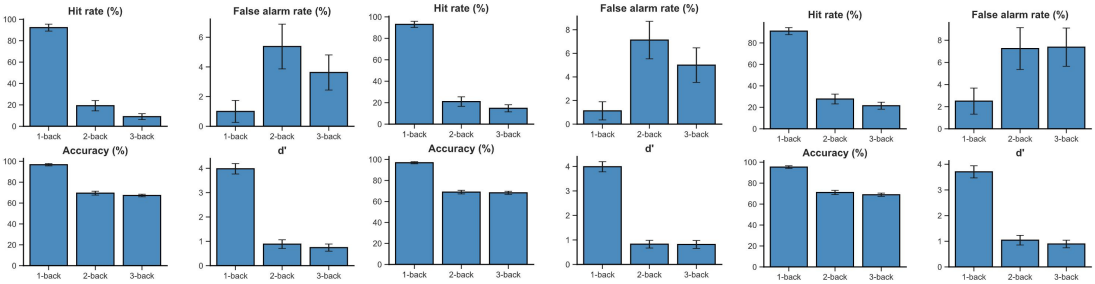

Figure 1: Metrics performance between different temperatures

6 Debug Diary: Main Blockers + How I Resolved Them

During the experiments, I encountered two main technical challenges that required code modifications.

The first issue occurred when running the high temperature condition (1.5). The model frequently produced invalid responses—instead of outputting just "m" or "-" as instructed, it sometimes generated random text, code snippets, or explanatory sentences. This caused the original response parsing logic to fail and raised exceptions, interrupting the experiment. To resolve this, I enhanced the response parsing with multiple fallback layers: first checking for exact matches, then cleaning and checking the first character, and finally searching within the response for 'm' or '-'. If still no valid response was found, I marked the trial as incorrect but allowed the program to continue execution rather than raising an error. This ensured the experiment ran to completion even with imperfect model outputs.

The second issue was the risk of data loss from interrupted runs. The original code only saved data at the very end of three n-back condition. Given that each temperature condition required 3,600 API calls and several hours to complete, any interruption due to API errors or network issues would have meant losing all progress. I solved this by implementing incremental saving after each block (50 blocks per n-back). I also added timestamps to filenames to avoid overwriting previous runs. This way, if an interruption occurred, only the current block's data would be lost, and the experiment could be resumed from the last saved point.

7 Conclusions: What Is Reproducible, What Isn't, and Why

7.1 What Is Reproducible

The core scientific finding of the original paper is fully reproducible: ChatGPT exhibits a human-like working memory capacity limit, with performance declining as

task difficulty increases. My results show the same qualitative pattern—d' decreases monotonically from n=1 to n=3 across all temperature conditions. This confirms that the capacity limit is a robust phenomenon, not an artifact of specific experimental conditions.

The relationships between the four metrics (hit rate, false alarm rate, accuracy, and d') also reproduce as expected. Higher d' corresponds to better discrimination ability, and the trade-off between hits and false alarms follows signal detection theory principles.

7.2 What Is Not Fully Reproducible

The exact quantitative values differ from the original paper. My baseline results show higher performance at n=1 (d'=3.99 vs. 3.53) but lower performance at n=2 (d'=0.83 vs. 1.43) and n=3 (d'=0.82 vs. 1.11). Consequently, the capacity limit in my replication appears at n=2 rather than n=3 as reported in the original study.

The response strategy also differs. My model becomes more liberal (higher false alarm rate) at n=2, while the original paper's model appears more controlled. This suggests changes in the model's default response bias.

Why These Differences Occur

Since I used the exact same letter sequences and same codebase as the original study, the differences cannot be attributed to experimental design. Three factors likely explain the discrepancies:

First, model version changes. OpenAI continuously updates gpt-3.5-turbo. The version I accessed in 2024 likely differs from the version used in the original 2023 study. These undocumented updates may affect working memory performance, particularly at higher cognitive loads.

Second, stochastic variation. Even at temperature=1.0, LLM outputs have inherent randomness. My results represent one sample from a distribution of possible outcomes. Multiple runs might show a range of d' values.

Third, API infrastructure changes. Even with identical code, underlying API routing, load balancing, or inference optimizations may have changed, introducing variability.

7.3 What My Modification Adds

My temperature modification revealed an important finding beyond reproduction: temperature effects are task-dependent. High temperature (1.5) improved performance on harder tasks (n=2 and n=3) but harmed performance on the easy task (n=1). This U-shaped effect resembles stochastic resonance in signal detection theory, suggesting that adding moderate randomness can help overcome systematic response biases when the model is uncertain. This finding has practical implications: optimal temperature settings may vary depending on task difficulty.

Despite quantitative differences, the core finding is robust: ChatGPT has limited working memory capacity. The reproducibility of this qualitative pattern across different runs and model versions suggests it is a fundamental property of the model architecture. My modification further shows that this limit can be partially modulated

by adjusting output randomness—a finding with both theoretical interest and practical applications.